MECH 325
Gear Stress



Objectives

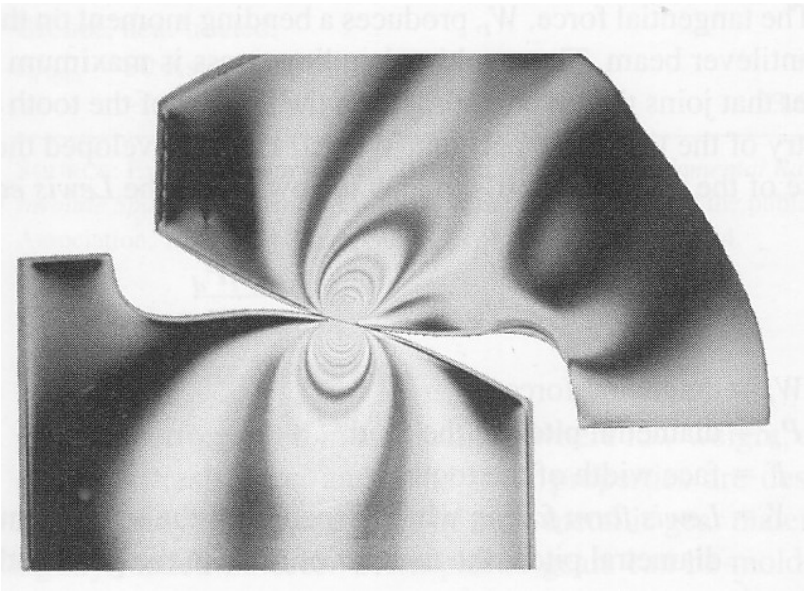
By the end of this section, you should be able to:

- Analyze bending and contact stress failure of spur and helical gears using AGMA standards

  - AGMA – American Gear Manufacturers Association

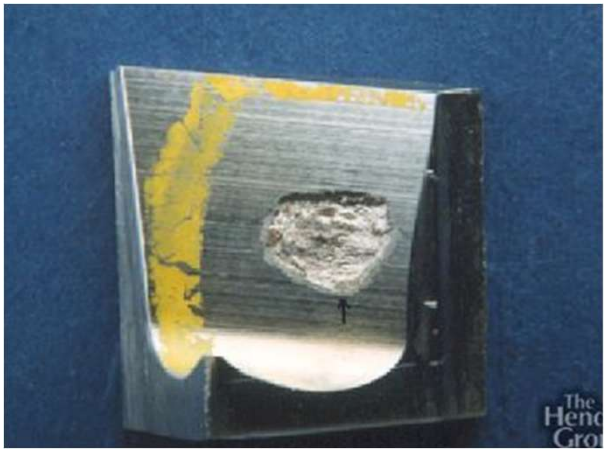
Stress in Gears



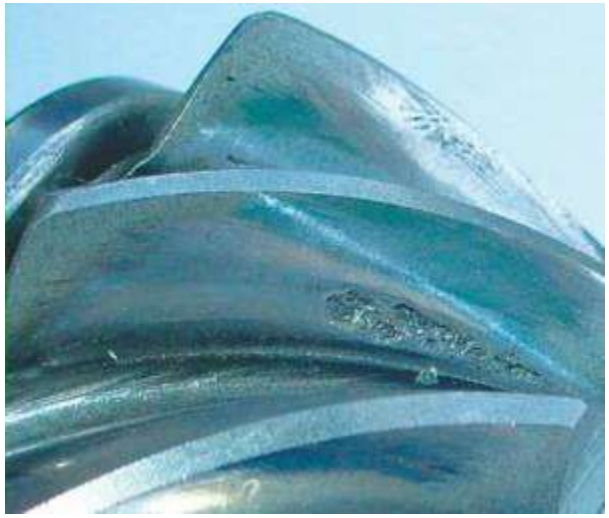


Stress Failures in Gears









AGMA Bending Stress Equations

Applied Loading Stress

$$\sigma = W^t K_o K_v K_s \frac{P_d}{F} \frac{K_m K_b}{J}$$

Material Strength

$$\sigma_{all} = \frac{S_t}{S_F} \frac{Y_N}{K_T K_R}$$

P_d is Diametrical Pitch (teeth/inch)
W^t (Shigley)= W_t (Mott) = Tangential Force



Bending stress safety factor

$$S_F = \frac{S_t}{\sigma} \frac{Y_N}{K_T K_R} = n_b$$

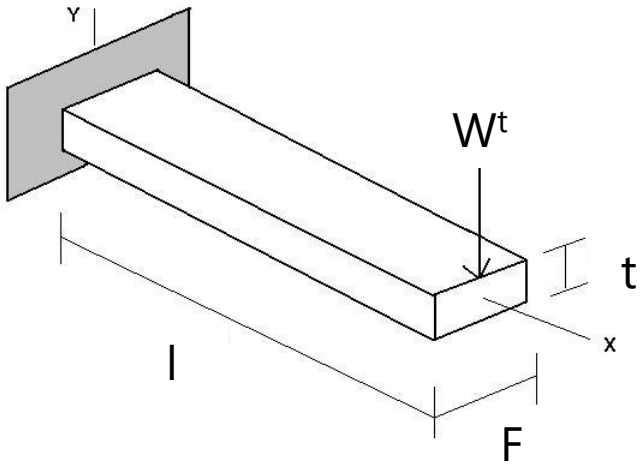
AGMA Empirical Factors

- Not all factors used in every case
- Refer to texts for complete details

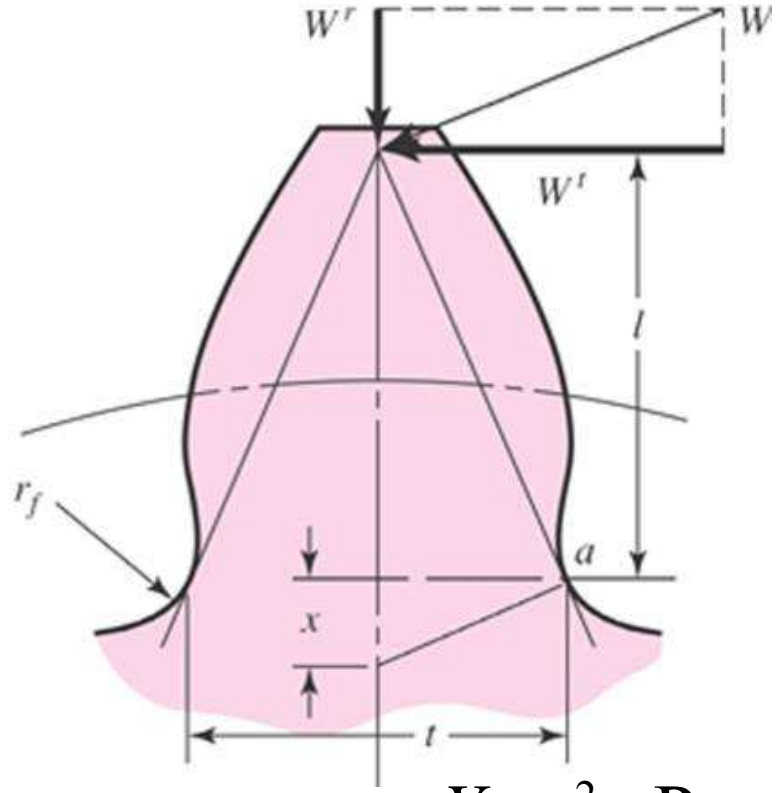
- C_f = surface condition factor
- C_H = hardness factor
- C_p = elastic coefficient
- I = geometry factor
- J = geometry factor
- K_B = rim thickness factor
- K_o = overload factor
- K_m = load distribution factor
- K_R = reliability factor
- K_s = size factor
- K_T = temperature factor
- K_v = dynamic factor
- Y_N = stress cycle factor

Lewis Equation for Gear Teeth (Bending Stress Equation)

Cantilever Beam Model



$$\sigma = \frac{6W^t l}{Ft^2}$$



$$Y = \tfrac{2}{3}\,\mathrm{xP}$$

(tabulated)

$$\sigma = \frac{W^t P}{FY}$$

Nominal Stress Value

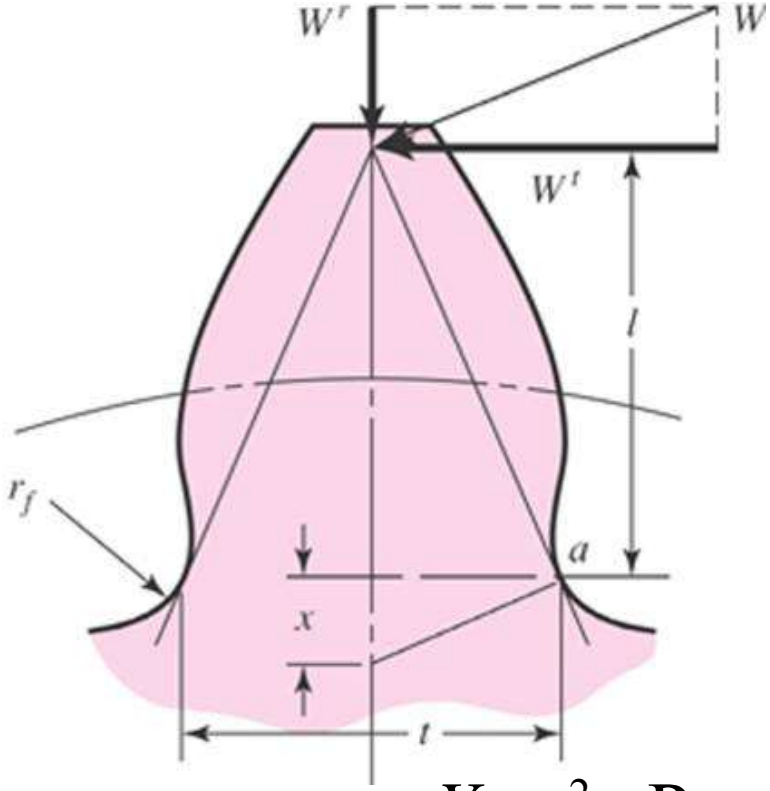
Table 14–1

Lewis Equation for Gear Teeth (Note: No longer used)

TABLE 9–15 Lewis Tooth Form Factor, *Y*, for Load Near the Pitch Point

Number Teeth	Tooth Form: 14 ½° Full Depth	20° Full Depth	20° Stub
26	0.522	0.588	0.678
28	0.535	0.597	0.688
30	0.540	0.606	0.698
34	0.553	0.628	0.714
38	0.566	0.651	0.729
43	0.575	0.672	0.739
50	0.588	0.694	0.758
60	0.604	0.713	0.774
75	0.613	0.735	0.792
100	0.622	0.757	0.808
150	0.635	0.779	0.830
300	0.650	0.801	0.855
Rack	0.660	0.823	0.881

Source: DuPont Polymers, Wilmington, DE.



$$Y = \frac{2}{3}\text{xP}$$

(tabulated)

$$\sigma = \frac{W^t P}{FY}$$

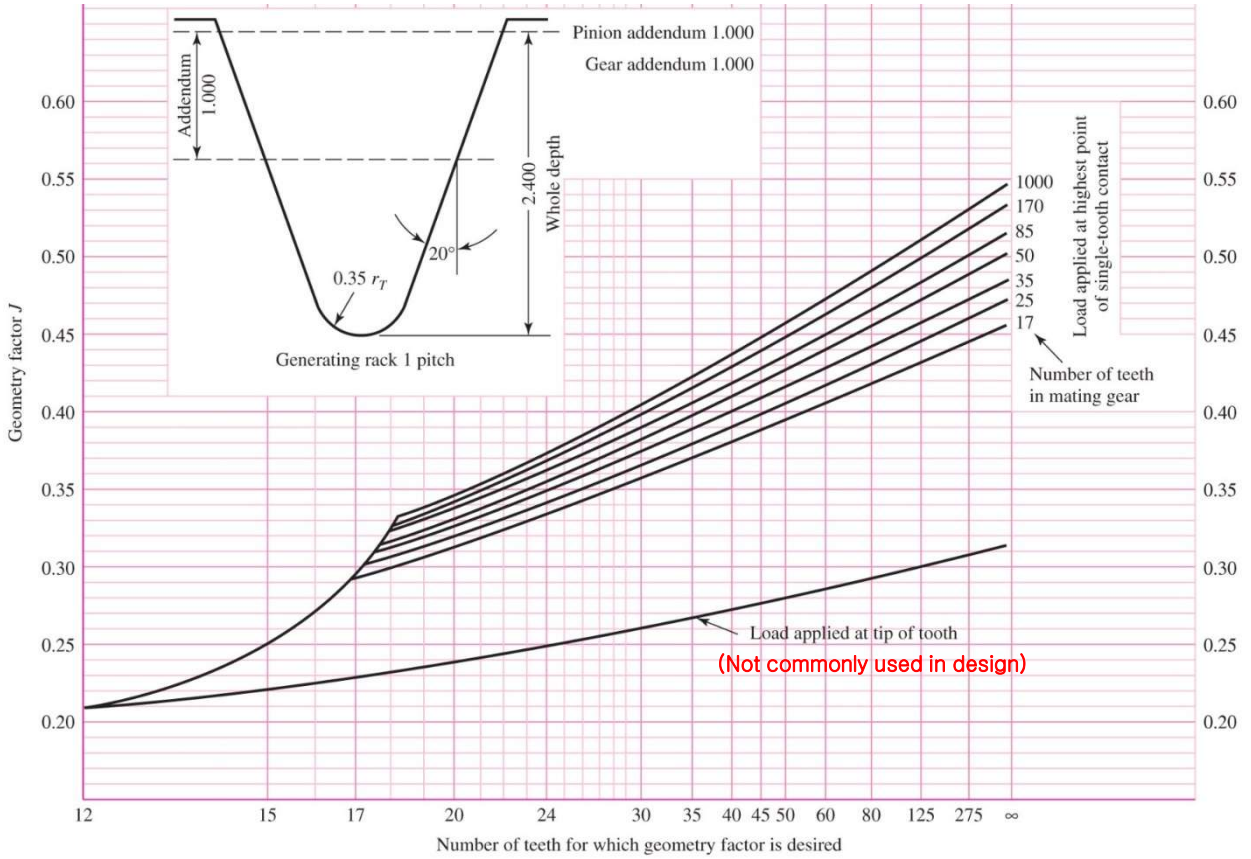
Nominal Stress Value

Table 14–1

Note: Form Factor Y is dimensionless

Geometry Factor *J*

- Accounts for shape of tooth in bending stress equation. It is related to the shape of the teeth and the fillet stress concentration A *nominal* bending stress is developed. This dimensionless value, J, replaced Lewis Form Factor, Y.
- The value of J increases with number of teeth as both the tooth profile and the load sharing ratio, m_c, increases..



Bending Stress Number, s_t

$$s_t = \frac{W_t P_d}{FJ} K_o K_s K_m K_B K_v \qquad (9\text{–}16)$$

where W_t = tangential force
P_d = diametral pitch
F = face width
J = geometry factor
K_o = overload factor for bending strength
K_s = size factor for bending strength
K_m = load-distrubution factor for bending strength
K_B = rim thickness factor
K_v = dynamic factor for bending strength

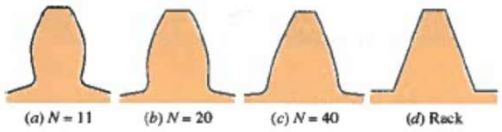


FIGURE 8–14 Involute curve shape for varying numbers of teeth for a diametral pitch of 5

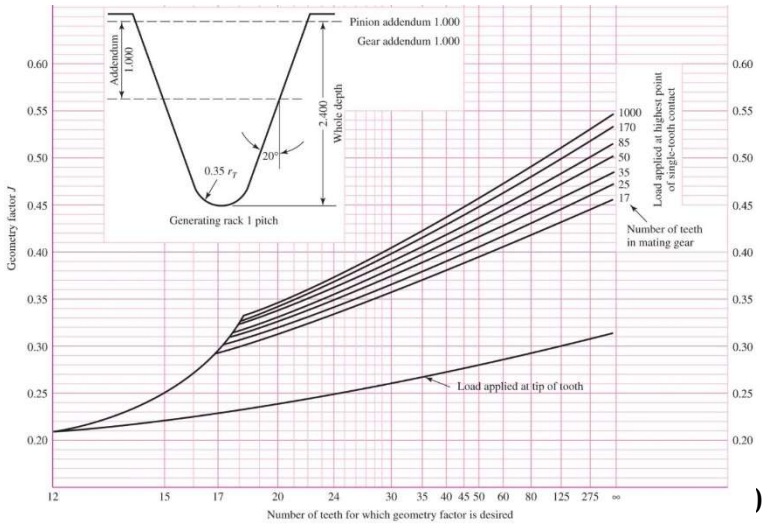
Bending Stress Number

- A *nominal stress number, s_t (Mott), σ (Shigley)* is calculated using the empirically derived J value.
- AGMA uses the term *stress number* instead of *stress* so that these values are not confused with the actual material stress properties (yield, ultimate, endurance, etc.)
- The nominal stress number is based on fatigue strength tests and not straight pull/push tests
- The geometry factor is conservative as it based on the location of the highest point of single contact and so it does not account for any load sharing.

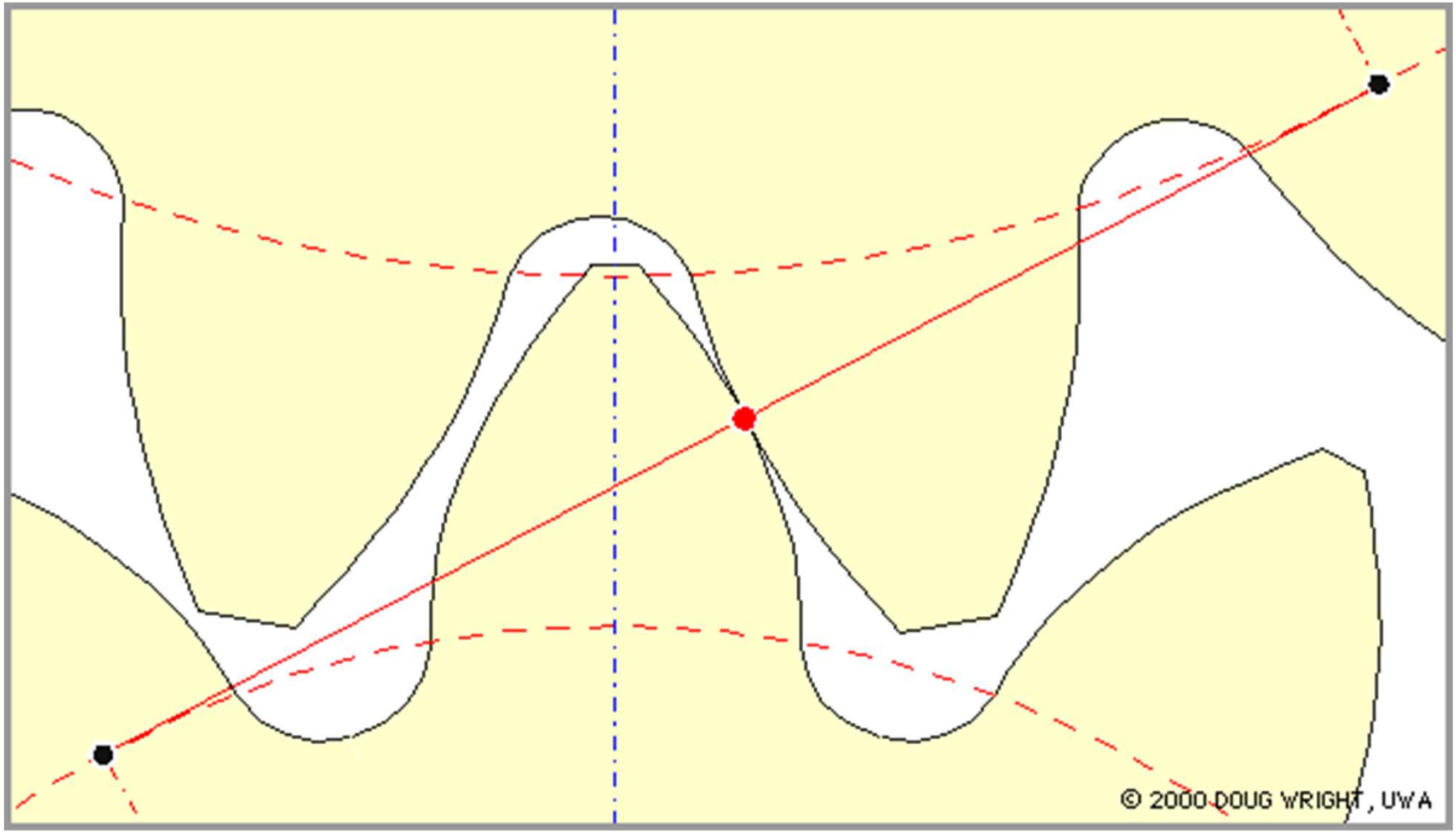
➲ **Bending Stress Number, s_t**

$$s_t = \frac{W_t P_d}{FJ} K_o K_s K_m K_B K_v \qquad (9\text{–}16)$$

where W_t = tangential force
P_d = diametral pitch
F = face width
J = geometry factor
K_o = overload factor for bending strength
K_s = size factor for bending strength
K_m = load-distrubution factor for bending strength
K_B = rim thickness factor
K_v = dynamic factor for bending strength



Involute Profile



http://www.mech.uwa.edu.au/DANotes/gears/meshing/gearAnimation.gif

Overload (Service) Factor K_o

- To account for likelihood of increase in nominal tangential load due to particular application.
- Recommended values,

Table of Overload Factors, K_o

Driven Machine			
Power source	Uniform	Moderate shock	Heavy shock
Uniform	1.00	1.25	1.75
Light shock	1.25	1.50	2.00
Medium shock	1.50	1.75	2.25

Shigley

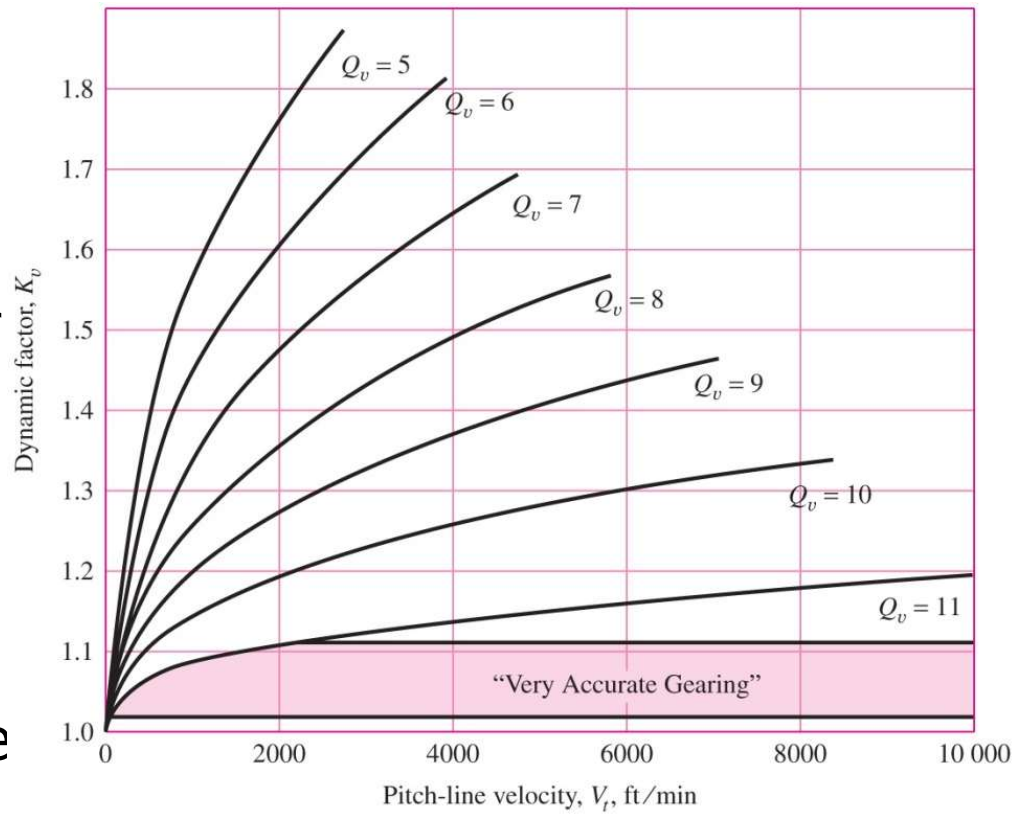
TABLE 9–1 Suggested Overload Factors, K_o

Driven Machine				
Power source	**Uniform**	**Light shock**	**Moderate shock**	**Heavy shock**
Uniform	1.00	1.25	1.50	1.75
Light shock	1.20	1.40	1.75	2.25
Moderate shock	1.30	1.70	2.00	2.75

Mott
(More Conservative)

Dynamic Factor K_v

- Dynamic factor K_v is a function of tangential speed, ωr, and gear quality Q.
- It accounts for impact of the teeth and the inaccuracies of the corresponding involute profiles.
- Other factors include the wear rates, friction, vibration, and imbalances



Dynamic Factor K_v

TABLE 9–6 Analytical Method for Computing Dynamic Factor, K_v

U.S. units		SI Metric units	
Given A_v and v_t (ft/min)		Given A_v and v_t (m/s)	
$B = 0.25(A_v - 5.0)^{0.667}$		$B = 0.25(A_v - 5.0)^{0.667}$	
$C = 50 + 56(1.0 - B)$		$C = 3.5637 + 3.9914(1.0 - B)$	
$K_v = \left[\frac{C}{C + \sqrt{v_t}}\right]^{-B}$		$K_v = \left[\frac{C}{C + \sqrt{v_t}}\right]^{-B}$	
Maximum velocity for a given A_v:		Maximum velocity for a given A_v:	
$v_{t\,max} = [C + (14 - A_v)]^2$		$v_{t\,max} = [C + (14 - A_v)]^2$	
A_v	$v_{t\,max}$ (ft/min)	A_v	$v_{t\,max}$ (m/s)
6	10 000	6	50.8
7	8239	7	41.9
8	6867	8	34.9
9	5731	9	29.1
10	4767	10	24.2
11	3937	11	20.0
12	3219	12	16.4

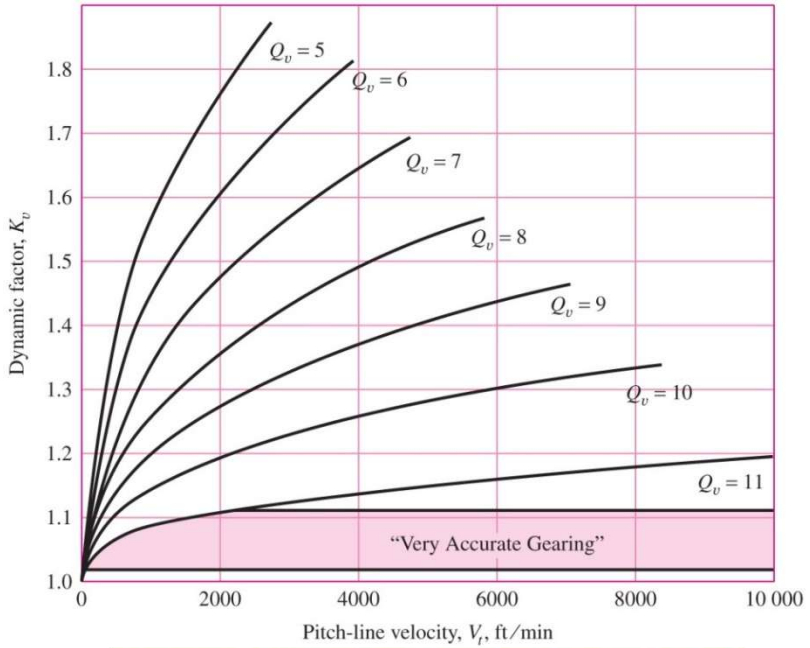


TABLE 9–4 General Corelations of AGMA 2000, AGMA 2015, and ISO 1328 Gear Quality Systems

AGMA 2000	AGMA 2015	ISO 1328	AGMA 2000	AGMA 2015	ISO 1328
—Least precise—					
Q5	—	12	Q11	A6	6
Q6	A11	11	Q12	A5	5
Q7	A10	10	Q13	A4	4
Q8	A9	9	Q14	A3	3
Q9	A8	8	Q15	A2	2
Q10	A7	7	—Most precise—		

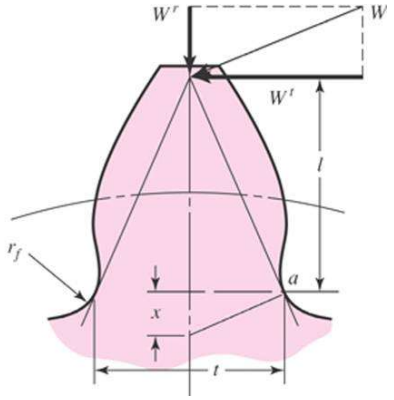
Note: The AGMA Gear Quality standards used in Shigley are no longer in use. The standards now correlate with ISO.

Size Factor K_s

- Same idea as the MARIN factor for size used to modify endurance strength
- The idea is "larger sizes" have higher probability for internal stress raisers.

- Size factor K_s
- Typically $K_s = 1$
- Or use

$$K_s = \frac{1}{k_b} = 1.192\left(\frac{F\sqrt{Y}}{P}\right)^{0.0535}$$



$$Y = \frac{2}{3}\text{xP}$$

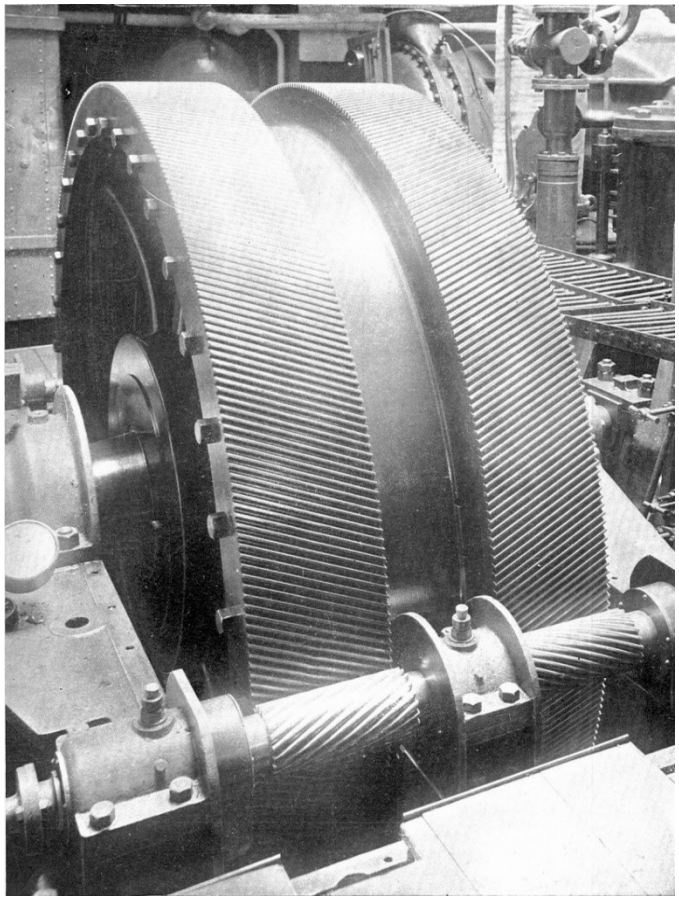
(tabulated)

TABLE 9–2 Suggested Size Factors, K_s

Diametral pitch, P_d	Metric module, m	Size factor, K_s
≥ 5	≤ 5	1.00
4	6	1.05
3	8	1.15
2	12	1.25
1.25	20	1.40

Load-Distribution Factor K_m

- Accounts for non-uniform distribution of load across the line of contact.
- Depends on mounting and face width.
- Especially important for large gears where alignment is critical.



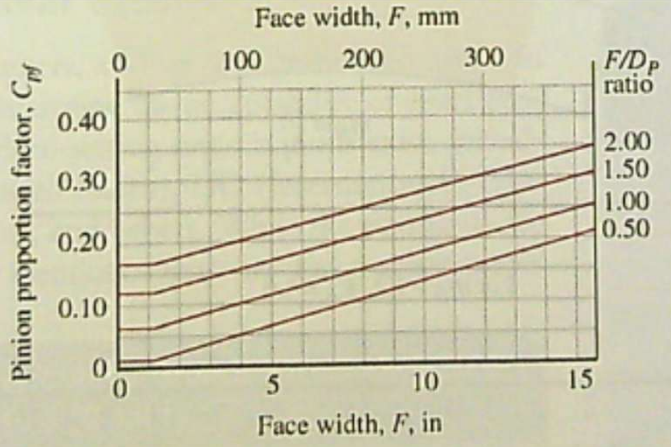
Load-Distribution Factor K_m (Mott)

- Two elements contribute the load distribution factor.
- C_{pf} (pinion face) is the pinion proportion factor (relative width of pinion)

$$K_m = 1.0 + C_{pf} + C_{ma} \qquad (9\text{–}17)$$

where C_{pf} = pinion proportion factor (see Figure 9–12)

C_{ma} = mesh alignement factor (see Figure 9–13)



D_P = Pinion diameter

For $F/D_P < 0.50$, use curve for $F/D_P = 0.50$

When $F \le 1.0$ in. ($F \le 25$ mm)

$$C_{pf} = \frac{F}{10D_P} - 0.025$$

When $1.0 \le F < 15$,

$$C_{pf} = \frac{F}{10D_P} - 0.0375 + 0.0125F$$

FIGURE 9–12 Pinion proportion factor, C_{pf} (Extracted from AGMA 2001 Standard, *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, with permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314)

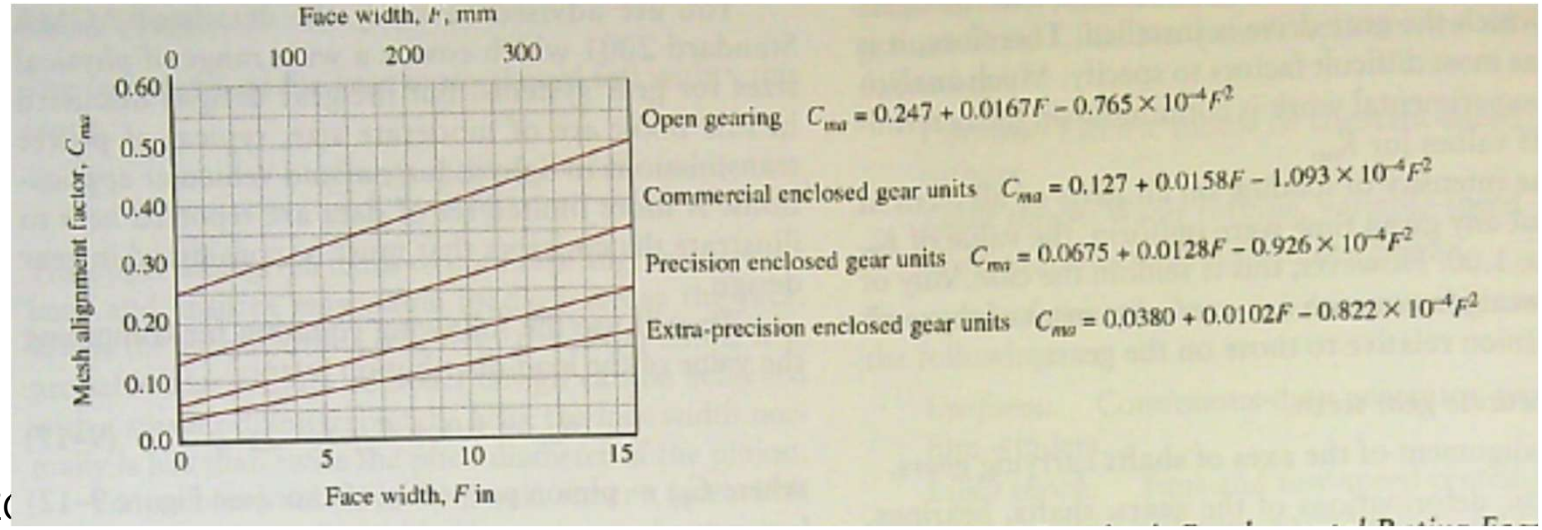
Load-Distribution Factor K_m (Mott)

- Two elements contribute the load distribution factor.
- C_{ma} (mesh alignment) defines the overall tolerance of the enclosure to house the gear drive
- Normally you can use the "Commercial enclosed gear unit" as the normal selection for C_{ma}.

$$K_m = 1.0 + C_{pf} + C_{ma} \qquad (9\text{–}17)$$

where C_{pf} = pinion proportion factor (see Figure 9–12)

C_{ma} = mesh alignement factor (see Figure 9–13)



Load-Distribution Factor K_m (Shigley)

- In this version the full parameters that define K_m are used

$$K_m = C_{mf} = 1 + C_{mc}(C_{pf}C_{pm} + C_{ma}C_e) \qquad (14\text{–}30)$$

$$C_{mc} = \begin{cases} 1 & \text{for uncrowned teeth} \\ 0.8 & \text{for crowned teeth} \end{cases} \qquad (14\text{–}31)$$

Normally uncrowned teeth are used

$$C_{pf} = \begin{cases} \dfrac{F}{10d_P} - 0.025 & F \le 1 \text{ in} \\ \dfrac{F}{10d_P} - 0.0375 + 0.0125F & 1 < F \le 17 \text{ in} \\ \dfrac{F}{10d_P} - 0.1109 + 0.0207F - 0.000\,228F^2 & 17 < F \le 40 \text{ in} \end{cases} \qquad (14\text{–}32)$$

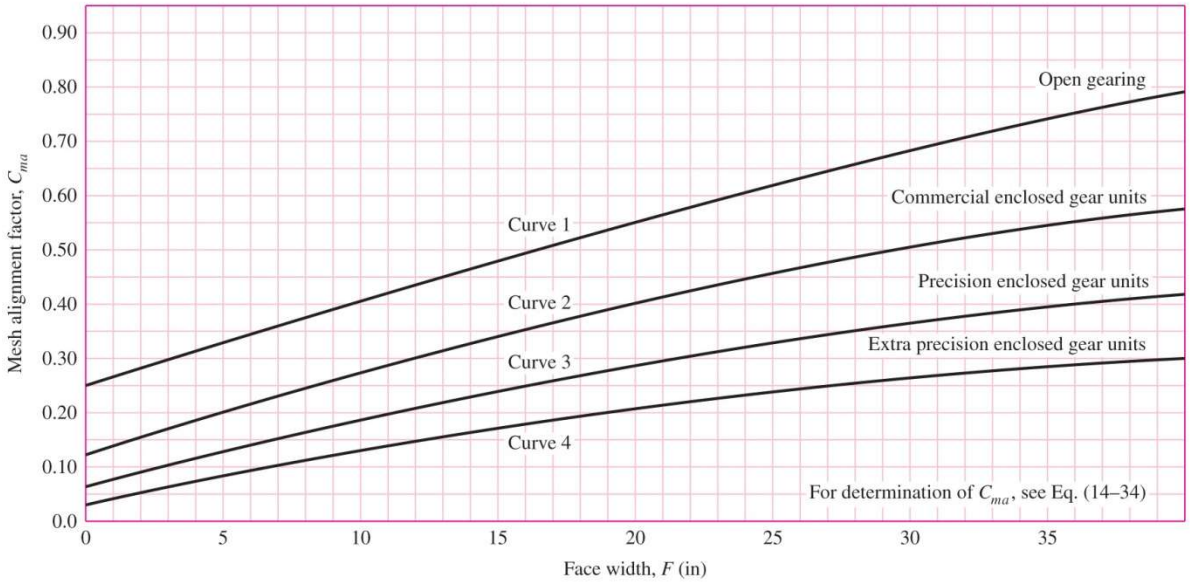
$$C_e = \begin{cases} 0.8 & \text{for gearing adjusted at assembly, or compatibility is improved by lapping, or both} \\ 1 & \text{for all other conditions} \end{cases} \qquad (14\text{–}35)$$

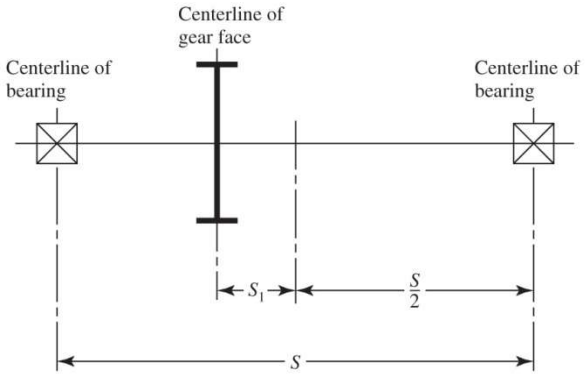
Normally 1.0 for commercial gears

Load-Distribution Factor K_m

$$K_m = C_{mf} = 1 + C_{mc}(C_{pf}C_{pm} + C_{ma}C_e) \qquad (14\text{–}30)$$

$$C_{pm} = \begin{cases} 1 & \text{for straddle-mounted pinion with } S_1/S < 0.175 \\ 1.1 & \text{for straddle-mounted pinion with } S_1/S \geq 0.175 \end{cases} \qquad (14\text{–}33)$$

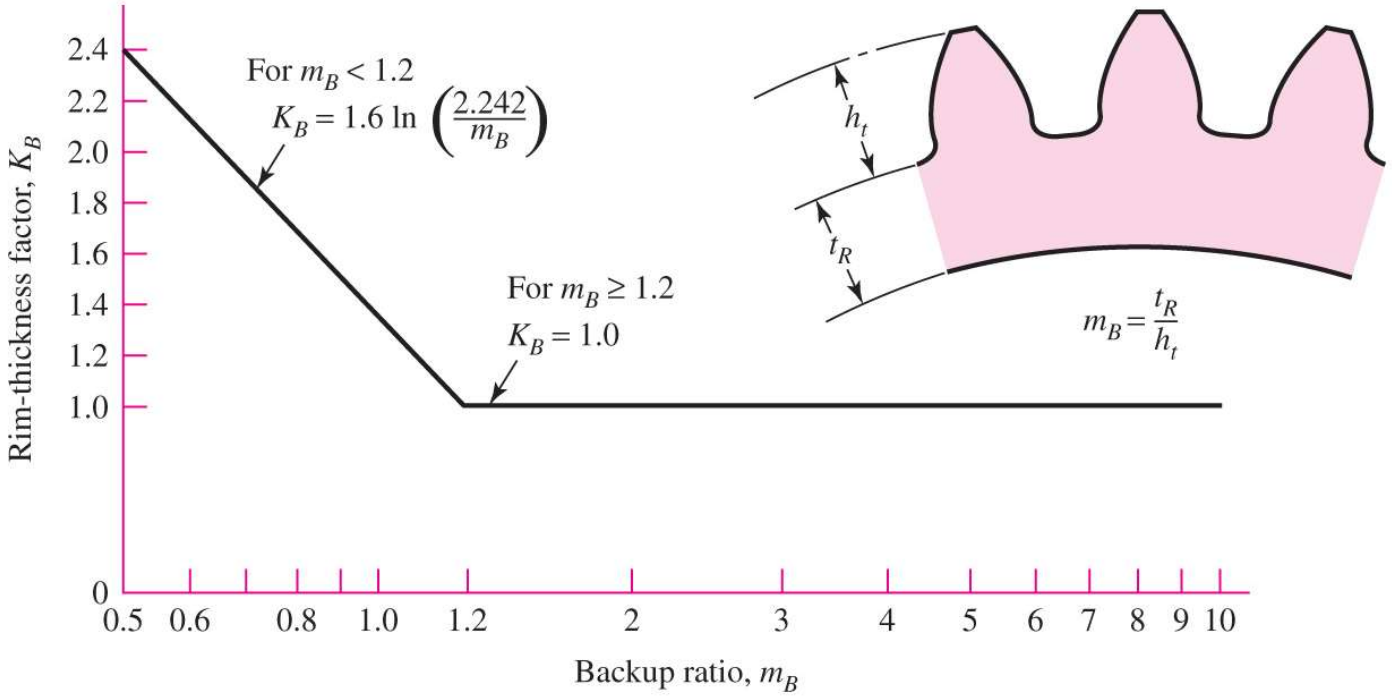




Unless other information is given, default to curve 2 (commercial enclosed gear units)

Rim-Thickness Factor K_B

- Accounts for bending of rim on a gear that is not solid
- Normally $K_B = 1.0$



SPUR GEAR BENDING FATIGUE
Based on ANSI/AGMA 2001-D04 (U.S. customary units)

$$d = \frac{N}{P}$$

$$V = \frac{\pi d n}{12} \quad \text{Eq. (13–34)}$$

$$W^t = \frac{33\,000\,H}{V} \quad \text{Eq. (13–35)}$$

1 [or Eq. (14–29)]

Transverse diametral pitch; a primary design parameter

Eq. (14–30)

Eq. (14–40)

Bending stress equation Eq. (14–15)

$$\sigma_b = W^t K_o K_s K_s \frac{P}{F} \frac{K_m K_B}{J}$$

Fig. 14–6

Face width; Typically, $\frac{3\pi}{P} < F < \frac{5\pi}{P}$

Eq. (14–27)

Table 14–9

Tables 14–3, 14–4

Allowable bending stress equation Eq. (14–17)

$$\sigma_{b,\text{all}} = \frac{S_t}{S_F} \frac{Y_N}{K_T K_R}$$

Fig. 14–14

Table 14–10, Eq. (14–38)

Sec. 14–17

1 if $T < 250°F$

Bending fatigue factor of safety Eq. (14–41)

$$n_b = \frac{\sigma_{b,\text{all}}}{\sigma_b}$$

Source: Data from ANSI/AGMA 2001-D04.

Bending Stress Number, s_t

$$s_t = \frac{W_t P_d}{FJ} K_o K_s K_m K_B K_v \qquad (9\text{–}16)$$

where W_t = tangential force
P_d = diametral pitch
F = face width
J = geometry factor
K_o = overload factor for bending strength
K_s = size factor for bending strength
K_m = load-distrubution factor for bending strength
K_B = rim thickness factor
K_v = dynamic factor for bending strength

Note: W^t (Shigley) and W_t (Mott) are the same tangential (transverse) force.

SPUR GEAR BENDING FATIGUE
Based on ANSI/AGMA 2001-D04 (U.S. customary units)

$$d = \frac{N}{P}$$

$$V = \frac{\pi d n}{12} \quad \text{Eq. (13–34)}$$

$$W^t = \frac{33\,000\,\mathrm{H}}{V} \quad \text{Eq. (13–35)}$$

Bending stress equation Eq. (14–15):

$$\sigma_b = W^t K_o K_v K_s \frac{P}{F} \frac{K_m K_B}{J}$$

- W^t: Eq. (13–35)
- K_o: 1 [or Eq. (14–29)]
- K_v: Eq. (14–30)
- K_s: Eq. (14–40)
- P: Transverse diametral pitch; a primary design parameter
- K_m: Fig. 14–6
- K_B: Eq. (14–27)
- F: Face width; Typically, $\frac{3\pi}{P} < F < \frac{5\pi}{P}$
- J: Table 14–9

Allowable bending stress equation Eq. (14–17):

$$\sigma_{b,\mathrm{all}} = \frac{S_t}{S_F} \frac{Y_N}{K_T K_R}$$

- S_t: Tables 14–3, 14–4
- S_F: Sec. 14–17
- Y_N: Fig. 14–14
- K_T: 1 if $T < 250°\mathrm{F}$
- K_R: Table 14–10, Eq. (14–38)

Bending fatigue factor of safety Eq. (14–41):

$$n_b = \frac{\sigma_{b,\mathrm{all}}}{\sigma_b}$$

Source: Data from ANSI/AGMA 2001-D04.

Bending Stress Number, s_t

$$s_t = \frac{W_t P_d}{F J} K_o K_s K_m K_B K_v \qquad (9\text{–}16)$$

where W_t = tangential force
P_d = diametral pitch
F = face width
J = geometry factor
K_o = overload factor for bending strength
K_s = size factor for bending strength
K_m = load-distrubution factor for bending strength
K_B = rim thickness factor
K_v = dynamic factor for bending strength

We can now determine the Gear bending stress number.

This can be considered a "fully corrected nominal stress'

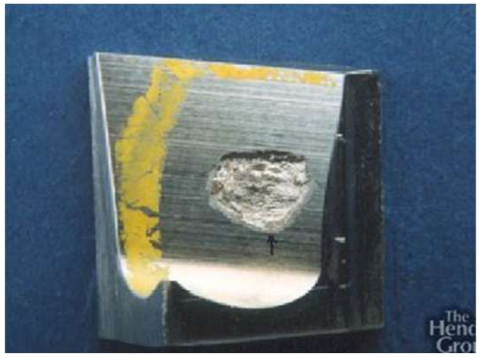
AGMA Contact Stress Equations (Pitting or Wear Stress)

$$\sigma_c = C_P \sqrt{W^t K_o K_v K_s \frac{K_m}{d_p F} \frac{C_f}{I}} \qquad \sigma_{c,all} = \frac{S_C}{S_H} \frac{Z_N C_H}{K_T K_R}$$

W^t (Shigley)= W_t (Mott) = Tangential Force

d_P is the Pitch Diameter (inches)

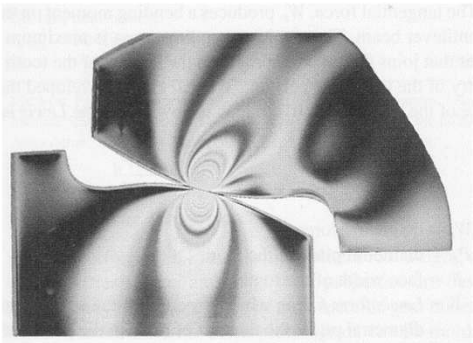
K_o K_v K_s K_m as before



Contact stress safety factor

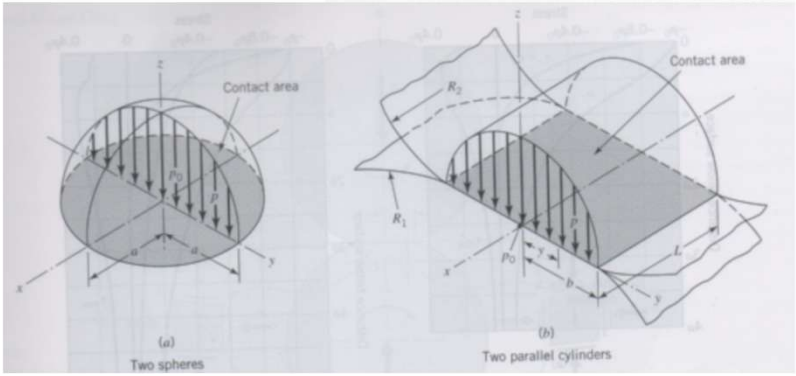
$$S_H = \frac{S_C}{\sigma_C} \frac{Z_N C_H}{K_T K_R} = \eta_C$$

Hertz Contact Stress



- When curved *elastic* bodies are in contact, *finite* contact areas are developed due to deflections in the materials.
- Compressive stresses are developed as a result of these deflections.
- The stress equations are credited to Heinrich Hertz (1881)





- The following stress equations are derived (F=Force, L= Face):

$$p_0 = 0.578\sqrt[3]{\frac{F(1/R_1 + 1/R_2)^2}{\Delta^2}}$$

Sphere Contact

$$a = 0.908\sqrt[3]{\frac{F\,\Delta}{1/R_1 + 1/R_2}}$$

$$p_0 = 0.564\sqrt{\frac{F(1/R_1 + 1/R_2)}{L\Delta}}$$

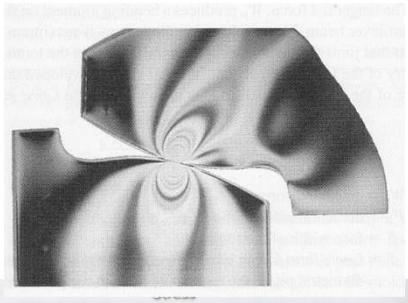
Cylinder Contact

$$b = 1.13\sqrt{\frac{F\Delta}{L(1/R_1 + 1/R_2)}}$$

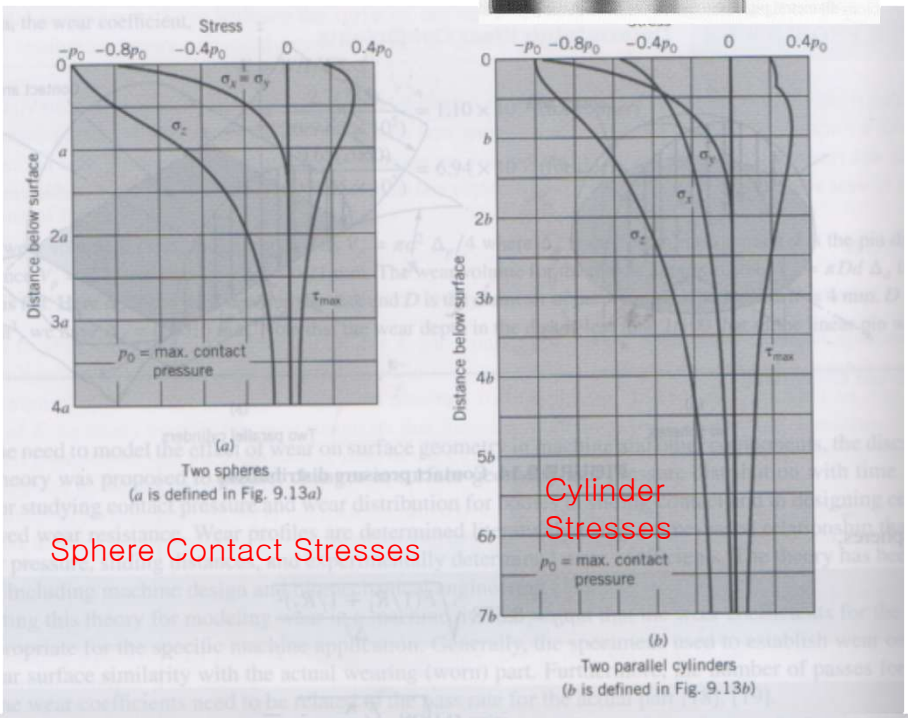
- The contact modulus, Δ, defined as;

$$\Delta = \frac{1-\upsilon_1^2}{E_1} + \frac{1-\upsilon_2^2}{E_2}$$

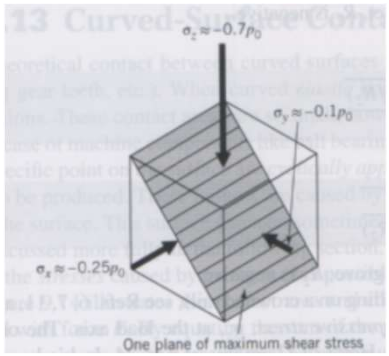
Hertz Contact Stress

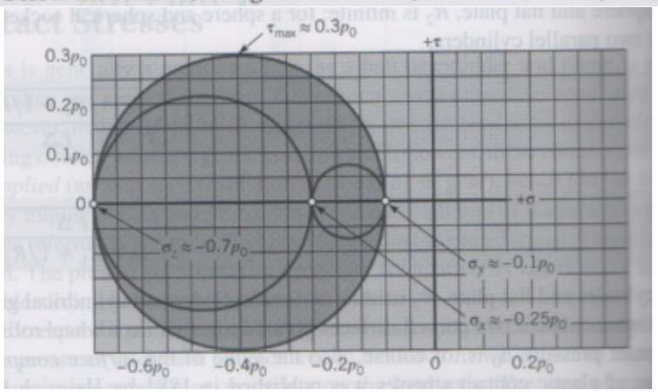


- Vertical contact stresses, σ_z, generate transverse compressive stresses, σ_x and σ_y due to Poissons ratio, υ.
- The three compressive stresses are the principal stresses for a stress element and can be plotted as a 3D Mohr Circle.
- Maximum shear stresses occur below the contact surface at a depth of approx. 0.5 the contact width, b. (Tresca Criteria).



Elastic stresses below the surface, along the load axis (the z–axis; $x = 0, y = 0$; for $\nu = 0.3$).

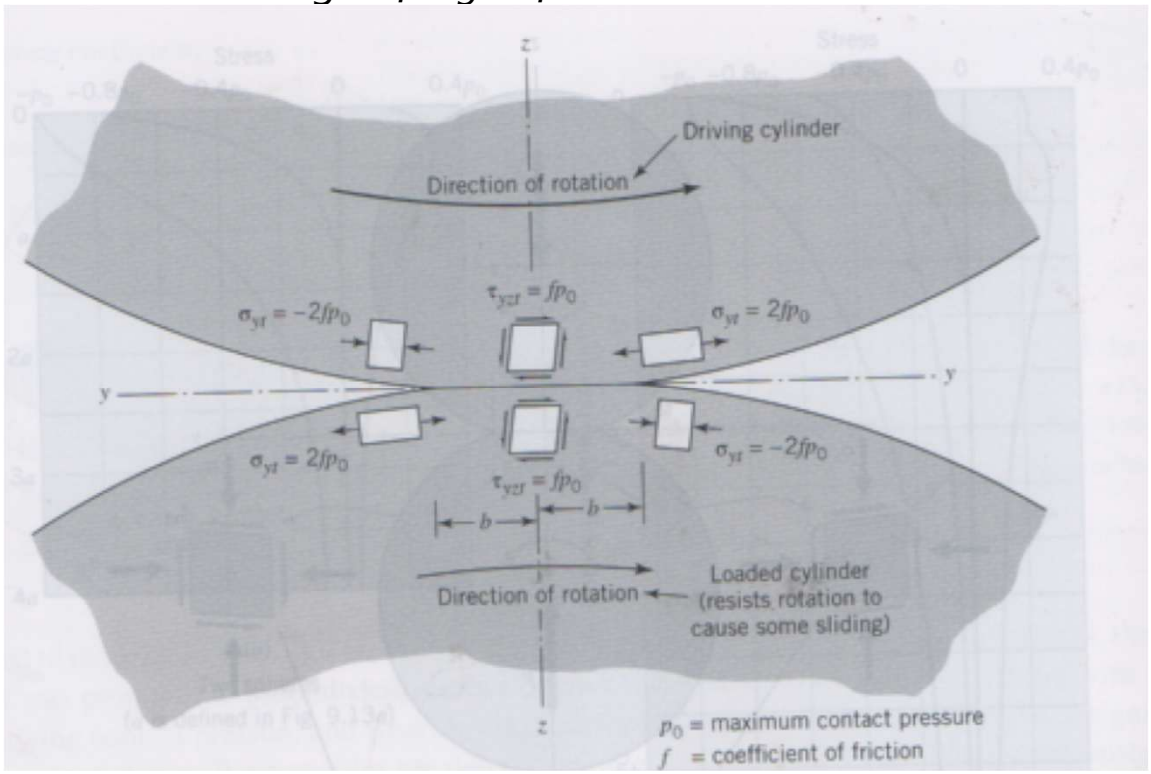




Rolling Friction Stress

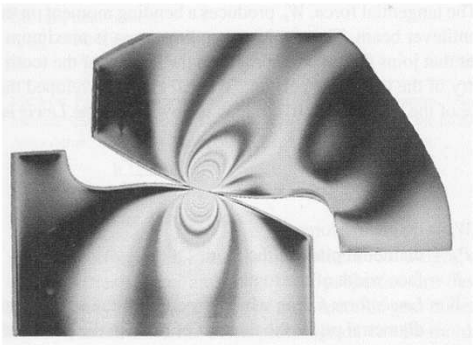
- The Hertz contact stresses are for the static load condtion. With the addition of motion along with friction and sliding, the Hertz contact stresses are not the only stress that must be modelled.
- Additional stresses include shear stresses due to friction and axial tension and compression stresses
- The stresses are cyclic in nature leading to *fatigue failure*.

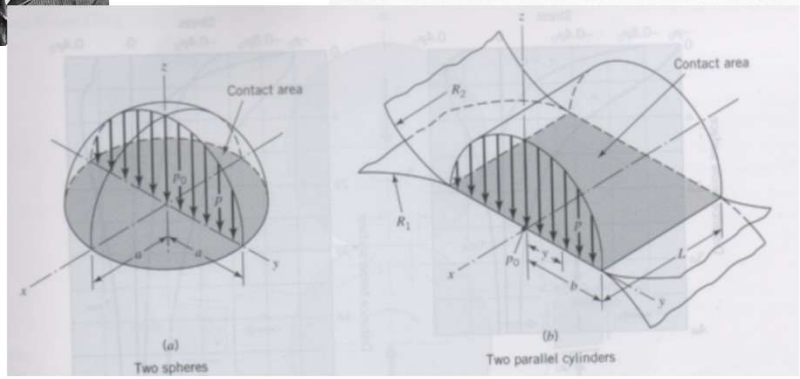




Hertz Contact Stress







- The contact stress equations are derived from the Hertz equations for Cylinder Contact.
- The involute profile provides an estimate for the radii required to derive the stress
- The contact load is the normal force acting on the pressure line. $W = W_t/\cos\phi$

$$\sigma_C = -\left[\frac{W^t}{\pi F \cos\phi}\,\frac{1/r_1 + 1/r_2}{(1-\nu_1^2)/E_1 + (1-\nu_2^2)/E_2}\right]^{1/2} \qquad \textbf{(14-11)}$$

Critical location is usually at the pitch line, where

$$r_1 = \frac{d_P \sin\phi}{2} \qquad r_2 = \frac{d_G \sin\phi}{2}$$

- The following stress equations are derived (F=Force, L= Face):

$$p_0 = 0.578\sqrt[3]{\frac{F(1/R_1 + 1/R_2)^2}{\Delta^2}}$$

Sphere Contact

$$a = 0.908\sqrt[3]{\frac{F\,\Delta}{1/R_1 + 1/R_2}}$$

$$p_0 = 0.564\sqrt{\frac{F(1/R_1 + 1/R_2)}{L\Delta}}$$

Cylinder Contact

$$b = 1.13\sqrt{\frac{F\Delta}{L(1/R_1 + 1/R_2)}}$$

- The contact modulus, Δ, defined as;

$$\Delta = \frac{1-\nu_1^2}{E_1} + \frac{1-\nu_2^2}{E_2}$$

Contact Stress Number

$$\sigma_c = C_P \sqrt{W^t K_o K_v K_s \frac{K_m}{d_p F} \frac{C_f}{I}}$$

- The nominal contact stress is modified in the same way as bending stress number.
  - K_o – overload factor
  - K_v - dynamic factor
  - K_s – size factor
  - K_m – load distribution factor
  - F – width of gear (face)
  - d_p - pitch diameter
  - Additional parameters – C_p, I, and C_f

Contact Stress Number
(9–23)

$$s_c = C_p \sqrt{\frac{W_t K_o K_s K_m K_v}{F D_P I}}$$

Mott Equation

Elastic Coefficient C_p

- This elastic coefficient is derived directly from the Hertz equation

$$\sigma_C = -\left[\frac{W^t}{\pi F \cos\phi}\,\frac{1/r_1 + 1/r_2}{(1-\nu_1^2)/E_1 + (1-\nu_2^2)/E_2}\right]^{1/2}$$

$$C_p = \left[\frac{1}{\pi\left(\frac{1-\nu_P^2}{E_P} + \frac{1-\nu_G^2}{E_G}\right)}\right]^{1/2}$$

- Values are found using the material properties for the contact materials

Table 14–8 Elastic Coefficient C_p (Z_E), $\sqrt{\text{psi}}$ ($\sqrt{\text{MPa}}$)

		Gear Material and Modulus of Elasticity E_G, lbf/in² (MPa)*					
Pinion Material	Pinion Modulus of Elasticity E_P psi (MPa)*	Steel 30 × 10^6 (2 × 10^5)	Malleable Iron 25 × 10^6 (1.7 × 10^5)	Nodular Iron 24 × 10^6 (1.7 × 10^5)	Cast Iron 22 × 10^6 (1.5 × 10^5)	Aluminum Bronze 17.5 × 10^6 (1.2 × 10^5)	Tin Bronze 16 × 10^6 (1.1 × 10^5)
Steel	30 × 10^6	2300	2180	2160	2100	1950	1900
	(2 × 10^5)	(191)	(181)	(179)	(174)	(162)	(158)
Malleable iron	25 × 10^6	2180	2090	2070	2020	1900	1850
	(1.7 × 10^5)	(181)	(174)	(172)	(168)	(158)	(154)
Nodular iron	24 × 10^6	2160	2070	2050	2000	1880	1830
	(1.7 × 10^5)	(179)	(172)	(170)	(166)	(156)	(152)
Cast iron	22 × 10^6	2100	2020	2000	1960	1850	1800
	(1.5 × 10^5)	(174)	(168)	(166)	(163)	(154)	(149)
Aluminum bronze	17.5 × 10^6	1950	1900	1880	1850	1750	1700
	(1.2 × 10^5)	(162)	(158)	(156)	(154)	(145)	(141)
Tin bronze	16 × 10^6	1900	1850	1830	1800	1700	1650
	(1.1 × 10^5)	(158)	(154)	(152)	(149)	(141)	(137)

Note the units for C_p

Normally we have steel to steel contact
C_p = 2300 psi^0.5 (191 Mpa^0.5)

Geometry Factor I

- AGMA developed a geometry factor for pitting to include the radii of curvature and the cos ϕ terms in the contact stress equation.
- The Geometry Factor requires information on gear ratio, number of pinion teeth and pressure angle.
- A number of equations are provided in Shigley to compute I.
- Mott provides graphs for gears with pressure angles of 20° and 25°

Called *pitting resistance geometry factor* by AGMA.

$$I = \begin{cases} \dfrac{\cos\phi_t \sin\phi_t}{2m_N} \dfrac{m_G}{m_G + 1} & \text{external gears} \\ \dfrac{\cos\phi_t \sin\phi_t}{2m_N} \dfrac{m_G}{m_G - 1} & \text{internal gears} \end{cases} \qquad \textbf{(14-23)}$$

$$m_G = \frac{N_G}{N_P} = \frac{d_G}{d_P} \qquad \textbf{(14-22)}$$

$$m_N = \frac{p_N}{0.95Z} \qquad \textbf{(14-21)}$$

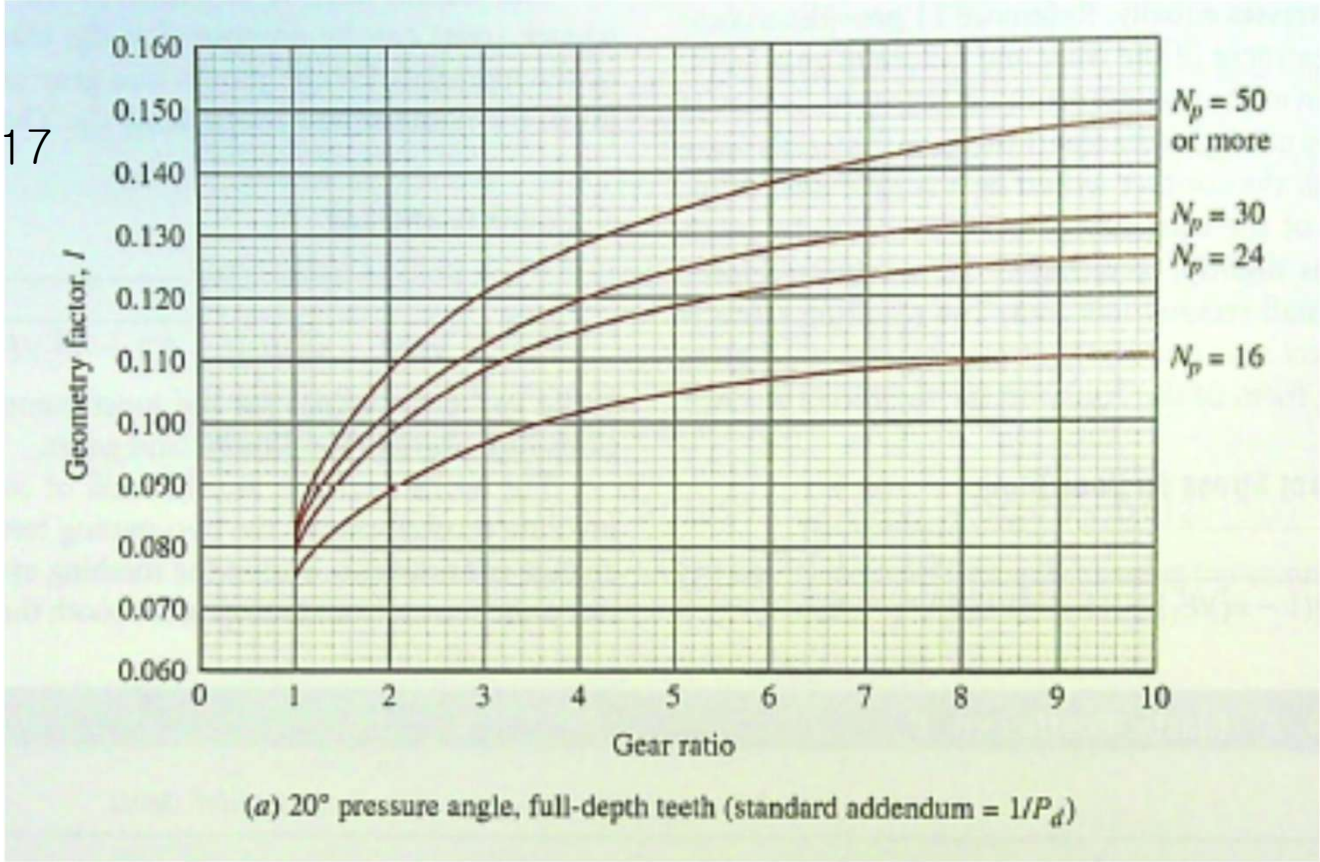
$$p_N = p_n \cos\phi_n \qquad \textbf{(14-24)}$$

$$Z = [(r_p + a)^2 - r_{bP}^2]^{1/2} + [(r_G + a)^2 - r_{bG}^2]^{1/2} - (r_P + r_G)\sin\phi_t \qquad \textbf{(14-25)}$$

$$r_b = r\cos\phi_t \qquad \textbf{(14-26)}$$

Geometry Factor I

Fig. 9–17



(a) 20° pressure angle, full-depth teeth (standard addendum = $1/P_d$)

Our nominal contact stress can now be computed $\sigma_c = C_p\sqrt{\frac{W_t}{FD_pI}}$

Surface Condition Factor, C_f

- To account for detrimental surface finish.
- No values currently given by AGMA.
- Use value of 1 for normal commercial gears.

$$\sigma_c = C_P \sqrt{W^t K_o K_v K_s \frac{K_m}{d_p F} \frac{C_f}{I}}$$

$$s_c = C_p \sqrt{\frac{W_t K_o K_s K_m K_v}{F D_P I}}$$

- C_f is not included in Mott
- We now have our fully modified contact stress number that we can compare with the material properties.

$$\sigma_{c,all} = \frac{S_C}{S_H} \frac{Z_N C_H}{K_T K_R}$$

Material Properties of Gear AGMA Allowable Strengths, S_t

- Gears must be made using high strength materials in order to accommodate the high stresses and wear
  - Through-hardened steels
  - Case-hardened steels
  - Flame- and induction-hardened gear teeth
  - Carburizing
  - Nitriding

- AGMA uses *allowable stress numbers* rather than *strengths*.
- The strength is normally defined by the Brinell Hardness Values

Material Strength

$$\sigma_{all} = \frac{S_t}{S_F}\frac{Y_N}{K_T K_R}$$

TABLE 9–8 Examples of Gear Materials

Heat treatment	Typical alloys (SAE numbers)
Through-hardened or Case-hardened by flame or induction hardening	1045, 4140, 4150, 4340, 4350
Carburizing, case-hardened	1020, 4118, 4320, 4820, 8620, 9310

AGMA Allowable Strengths, S_t

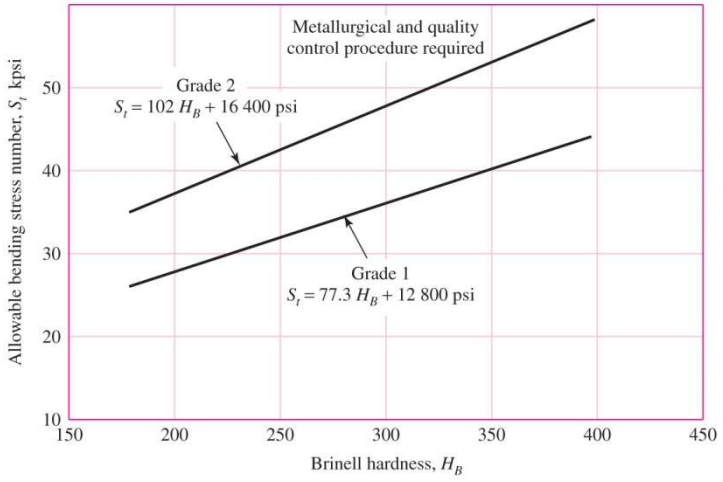
- AGMA uses *allowable stress numbers* rather than *strengths*.
- The gear strength values are only for use with the AGMA stress values, and should not be compared with other true material strengths.
- Different allowable stress numbers are provided for bending and contacts stresses
- Shigley 2024 replaced *Allowable Stress Numbers* with *Gear Bending (Contact) Stress*
- The strength values are defined for fatigue life (similar to endurance strength)

Table 14–3

Repeatedly Applied Bending Strength S_t at 10^7 Cycles and 0.99 Reliability for Steel Gears

Source: ANSI/AGMA 2001-D04.

Material Designation	Heat Treatment	Minimum Surface Hardness[1]	Allowable Bending Stress Number S_t,[2] psi		
			Grade 1	Grade 2	Grade 3
Steel[3]	Through-hardened	See Fig. 14–2	See Fig. 14–2	See Fig. 14–2	—
	Flame[4] or induction hardened[4] with type A pattern[5]	See Table 8*	45 000	55 000	—
	Flame[4] or induction hardened[4] with type B pattern[5]	See Table 8*	22 000	22 000	—
	Carburized and hardened	See Table 9*	55 000	65 000 or 70 000[6]	75 000
	Nitrided[4,7] (through-hardened steels)	83.5 HR15N	See Fig. 14–3	See Fig. 14–3	—
Nitralloy 135M, Nitralloy N, and 2.5% chrome (no aluminum)	Nitrided[4,7]	87.5 HR15N	See Fig. 14–4	See Fig. 14–4	See Fig. 14–4



Note: These values are only for bending

AGMA Allowable Strengths, S_t

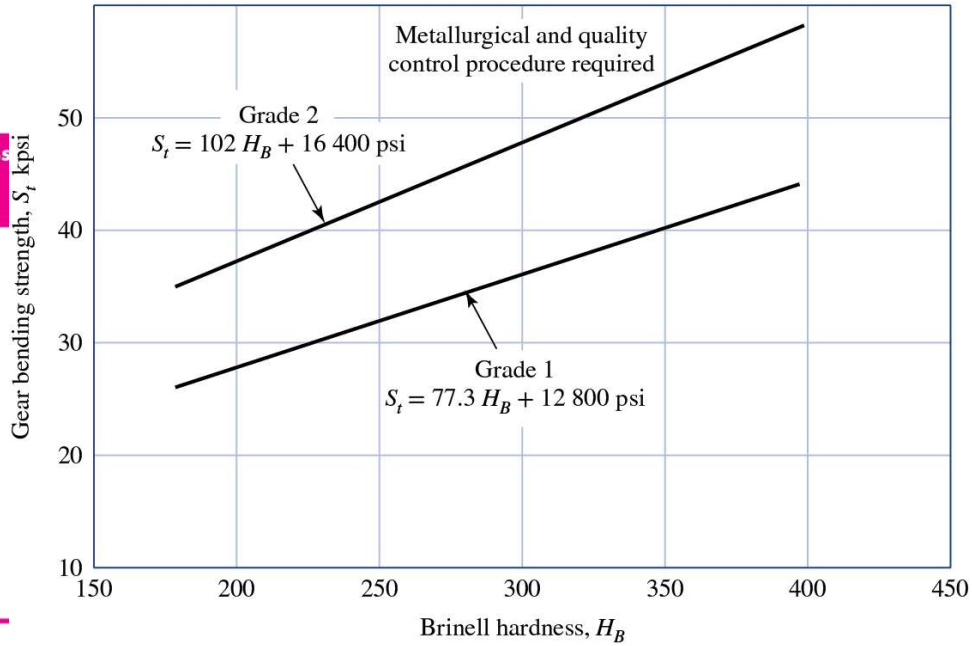
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Table 14–3

Repeatedly Applied Bending Strength S_t at 10^7 Cycles and 0.99 Reliability for Steel Gears

Source: ANSI/AGMA 2001-D04.

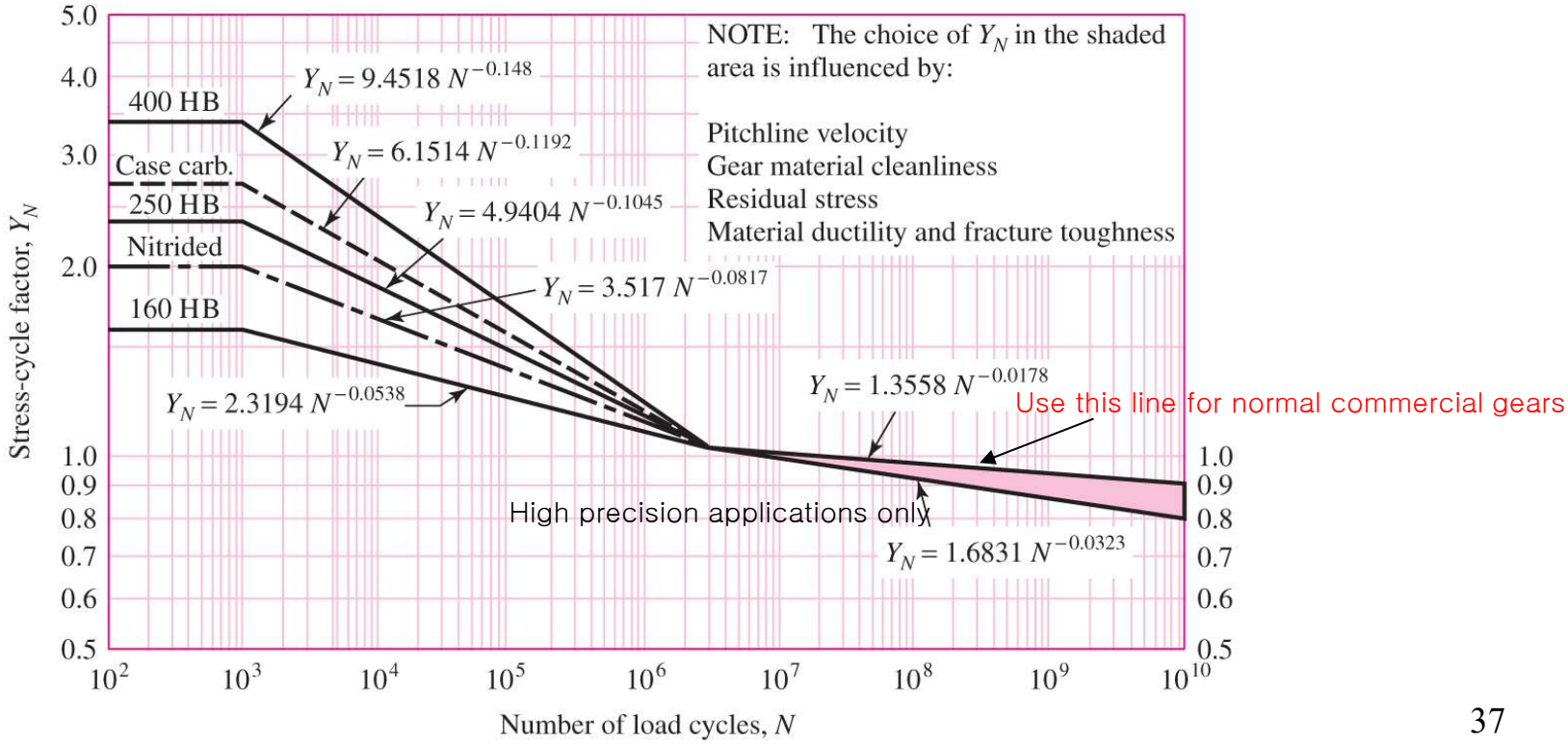
Material Designation	Heat Treatment	Minimum Surface Hardness[1]	Allowable Bending Stress, psi	
			Grade 1	Grade 2
Steel[3]	Through-hardened	See Fig. 14–2	See Fig. 14–2	See Fig. 14–2
	Flame[4] or induction hardened[4] with type A pattern[5]	See Table 8*	45 000	55 000
	Flame[4] or induction hardened[4] with type B pattern[5]	See Table 8*	22 000	22 000
	Carburized and hardened	See Table 9*	55 000	65 000 or 70 000[6]
	Nitrided[4,7] (through-hardened steels)	83.5 HR15N	See Fig. 14–3	See Fig. 14–3
Nitralloy 135M, Nitralloy N, and 2.5% chrome (no aluminum)	Nitrided[4,7]	87.5 HR15N	See Fig. 14–4	See Fig. 14–4



Note: These values are only for bending

Stress-Cycle Factors Y_N

- Similar to the fatigue S-N curve
- AGMA strengths are for 10^7 cycles
- Stress-cycle factors account for endurance of the gear train
- Fig. 14–14 gives Y_N for bending



Reliability Factor K_R (Y_Z)

- Accounts for statistical distributions of material fatigue failures
- Does not account for load variation
- Use Table 14–10 (Normally 99% is used K_R = 1.00)

Reliability	K_R (Y_Z)
0.9999	1.50
0.999	1.25
0.99	1.00
0.90	0.85
0.50	0.70

Temperature Factor K_T

- AGMA has not established values for this factor.
- For temperatures up to 250ºF (120ºC), $K_T = 1$ is acceptable.

SPUR GEAR CONTACT-STRESS FATIGUE
Based on ANSI/AGMA 2001-D04 (U.S. customary units)

$$d_P = \frac{N_P}{P}$$

$$V = \frac{\pi d n}{12} \quad \text{Eq. (13–34)}$$

$$W^t = \frac{33\,000\,\mathrm{H}}{V} \quad \text{Eq. (13–35)}$$

Contact stress equation Eq. (14–16)

$$\sigma_c = C_p \left(W^t K_o K_v K_s \frac{K_m}{d_P F} \frac{C_f}{I} \right)^{1/2}$$

Eq. (14–13), Table 14–8

1 [or Eq. (14–29)

Eq. (14–30)

1

Eq. (14–23)

Eq. (14–27)

Table 14–9

Allowable contact stress equation Eq. (14–18)

$$\sigma_{c,\mathrm{all}} = \frac{S_c Z_N C_H}{S_H K_T K_R}$$

Tables 14–6, 14–7

Fig. 14–15

Section 14–12, gear only

Table 14–10, Eq. (14–38)

Sec. 14–17

1 if $T < 250°\mathrm{F}$

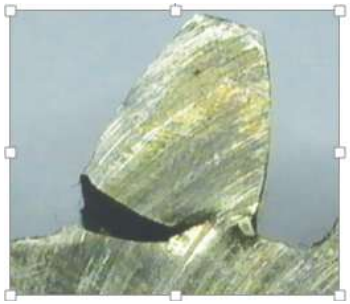
Contact-stress fatigue factor of safety Eq. (14–42)

$$n_c = \frac{\sigma_{c,\mathrm{all}}}{\sigma_c}$$

Safety Factor for Bending Stress, n_b , SF, S_F

- Included as design margins in the strength equations
- AGMA suggests SF of 1 to 1.5

$$S_F = \frac{S_t Y_N / (K_T K_R)}{\sigma} = \frac{\text{fully corrected bending strength}}{\text{bending stress}} \quad (\sigma_{all})$$



Bending stress safety factor

$$S_F = \frac{S_t}{\sigma} \frac{Y_N}{K_T K_R}$$

Bending fatigue factor of safety Eq. (14–41)

$$n_b = \frac{\sigma_{b,all}}{\sigma_b}$$

AGMA Allowable Strengths, S_c

- The AGMA *allowable stress numbers* are different between bending stress and contact stress.

Table 14–6 Repeatedly Applied Gear Contact Strength S_c at 10^7 Cycles and 0.99 Reliability for Steel Gears

Material Designation	Heat Treatment	Minimum Surface Hardness[1]	Gear Contact Strength,[2] S_c, psi		
			Grade 1	Grade 2	Grade 3
Steel[3]	Through-hardened[4]	See Fig. 14–5	See Fig. 14–5	See Fig. 14–5	—
	Flame[5] or induction hardened[5]	50 HRC 54 HRC	170 000 175 000	190 000 195 000	— —
	Carburized and hardened[5]	See Table 9*	180 000	225 000	275 000
	Nitrided[5] (through hardened steels)	83.5 HR15N 84.5 HR15N	150 000 155 000	163 000 168 00	175 000 180 000
2.5% chrome (no aluminum)	Nitrided[5]	87.5 HR15N	155 000	172 000	189 000
Nitralloy 135M	Nitrided[5]	90.0 HR15N	170 000	183 000	195 000
Nitralloy N	Nitrided[5]	90.0 HR15N	172 000	188 000	205 000
2.5% chrome (no aluminum)	Nitrided[5]	90.0 HR15N	176 000	196 000	216 000

Note: These values are only for contact stress numbers

AGMA Allowable Strengths, S_c

- The AGMA *allowable stress numbers* are different between bending stress and contact stress.

Table 14–6 Repeatedly Applied Gear Contact Strength S_c at 10^7 Cycles and 0.99 Reliability for Steel Gears

Material Designation	Heat Treatment	Minimum Surface Hardness[1]	Gear Contact Strength	
			Grade 1	Grade 2
Steel[3]	Through-hardened[4]	See Fig. 14–5	See Fig. 14–5	See Fig. 14–5
	Flame[5] or induction hardened[5]	50 HRC 54 HRC	170 000 175 000	190 000 195 000
	Carburized and hardened[5]	See Table 9*	180 000	225 000
	Nitrided[5] (through hardened steels)	83.5 HR15N 84.5 HR15N	150 000 155 000	163 000 168 00
2.5% chrome (no aluminum)	Nitrided[5]	87.5 HR15N	155 000	172 000
Nitralloy 135M	Nitrided[5]	90.0 HR15N	170 000	183 000
Nitralloy N	Nitrided[5]	90.0 HR15N	172 000	188 000
2.5% chrome (no aluminum)	Nitrided[5]	90.0 HR15N	176 000	196 000

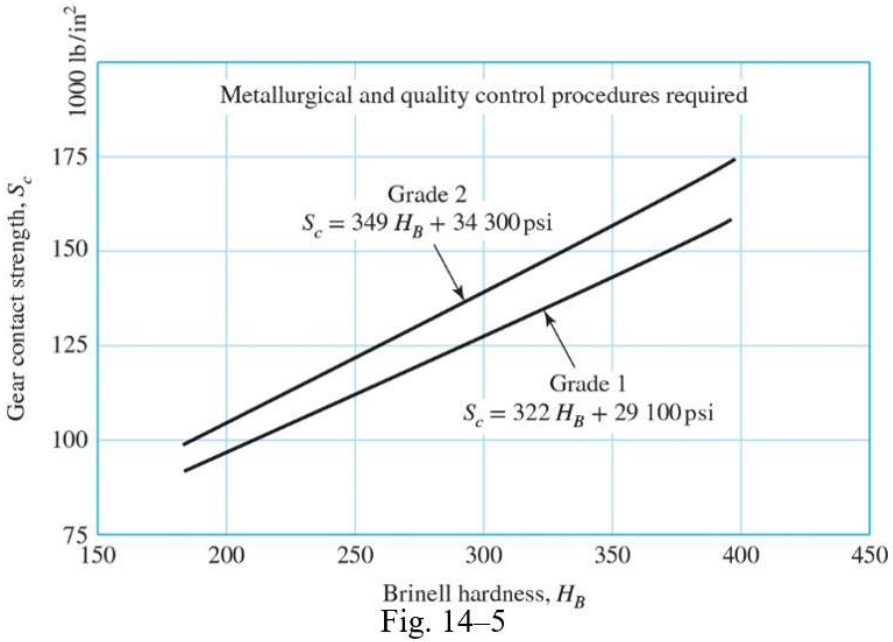


Fig. 14–5

Note: These values are only for contact stress numbers

Contact Stress Cycle Factor Z_n

- The same type of S-N style curve

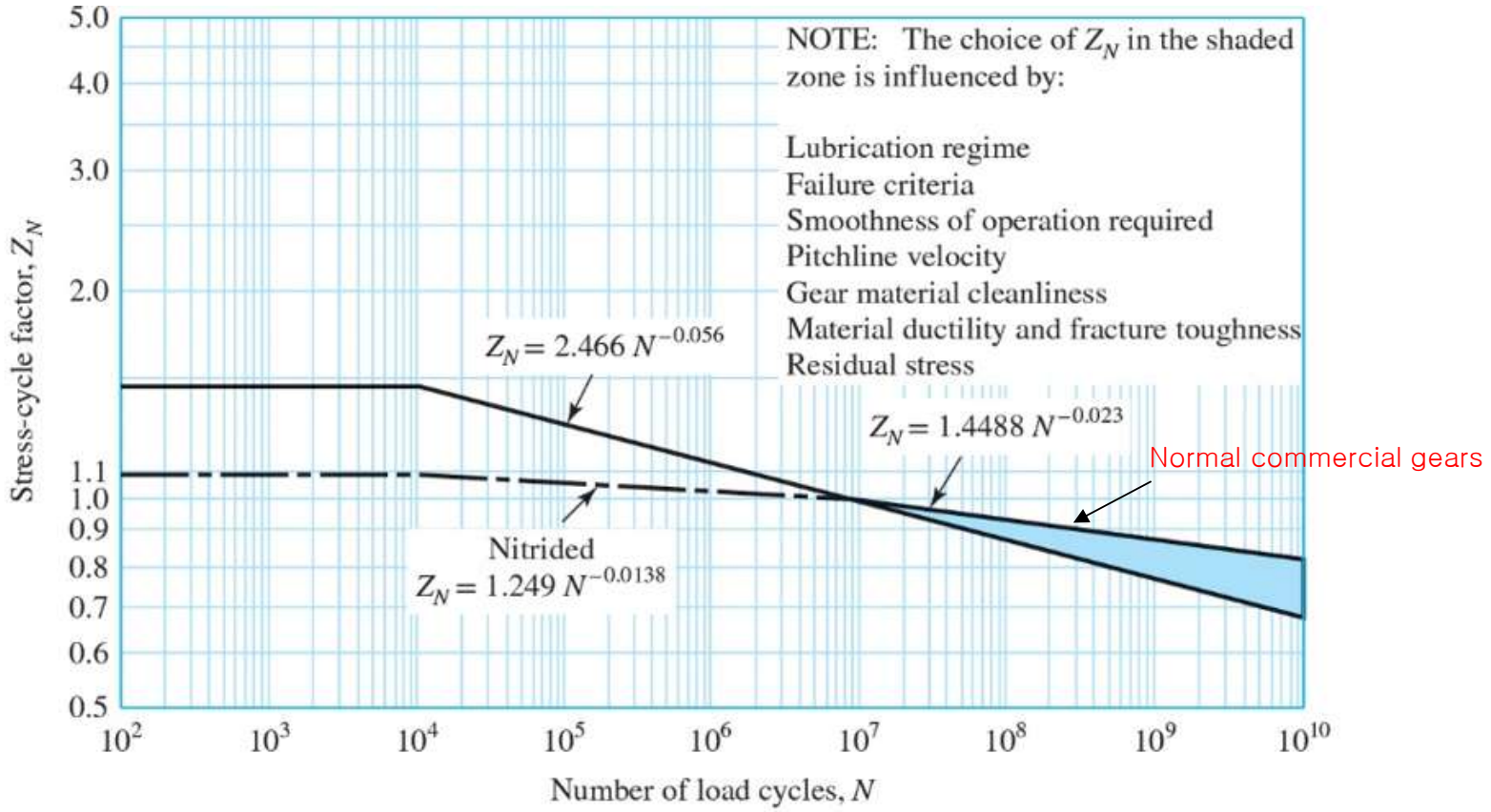


Fig. 14–15

Hardness-Ratio Factor C_H

- Since the pinion is subjected to more cycles than the gear, it is often hardened more than the gear.
- The hardness-ratio factor accounts for the difference in hardness of the pinion and gear.
- C_H is only applied to the gear.
- That is, C_H = 1 for the pinion.

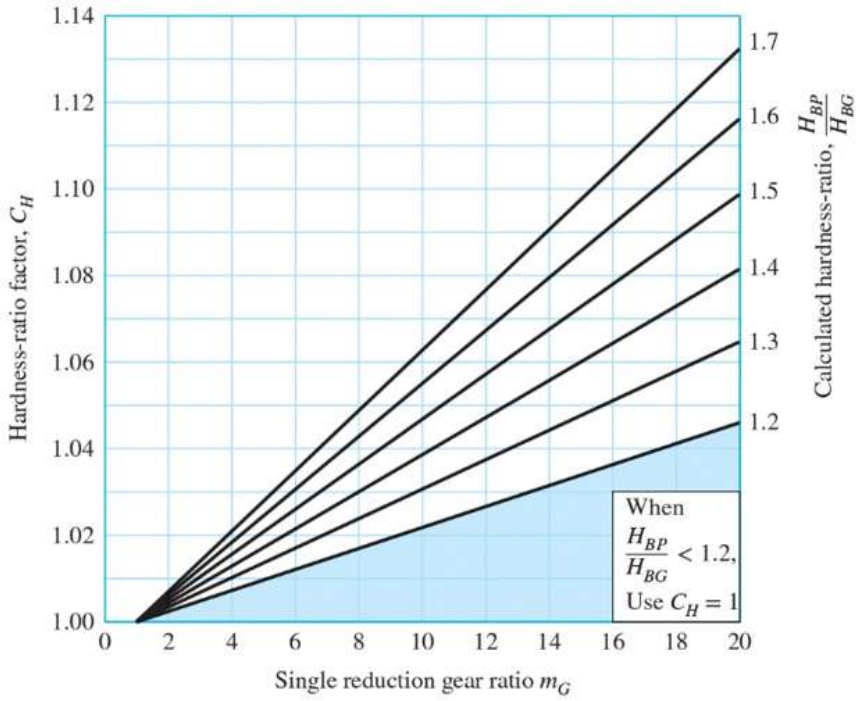
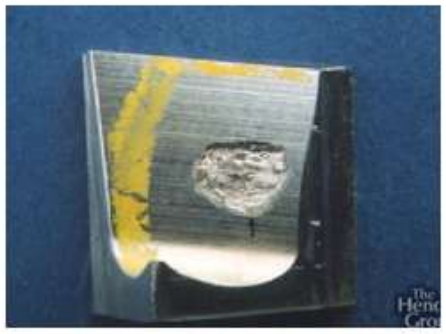


Fig. 14–12

Safety Factor for Contact Stress S_H

- Included as design margins in the contact strength equations
- AGMA suggests SF of 1 to 1.5

$$S_H = \frac{S_c Z_N C_H / (K_T K_R)}{\sigma_c} = \frac{\text{fully corrected contact strength}}{\text{contact stress}}$$

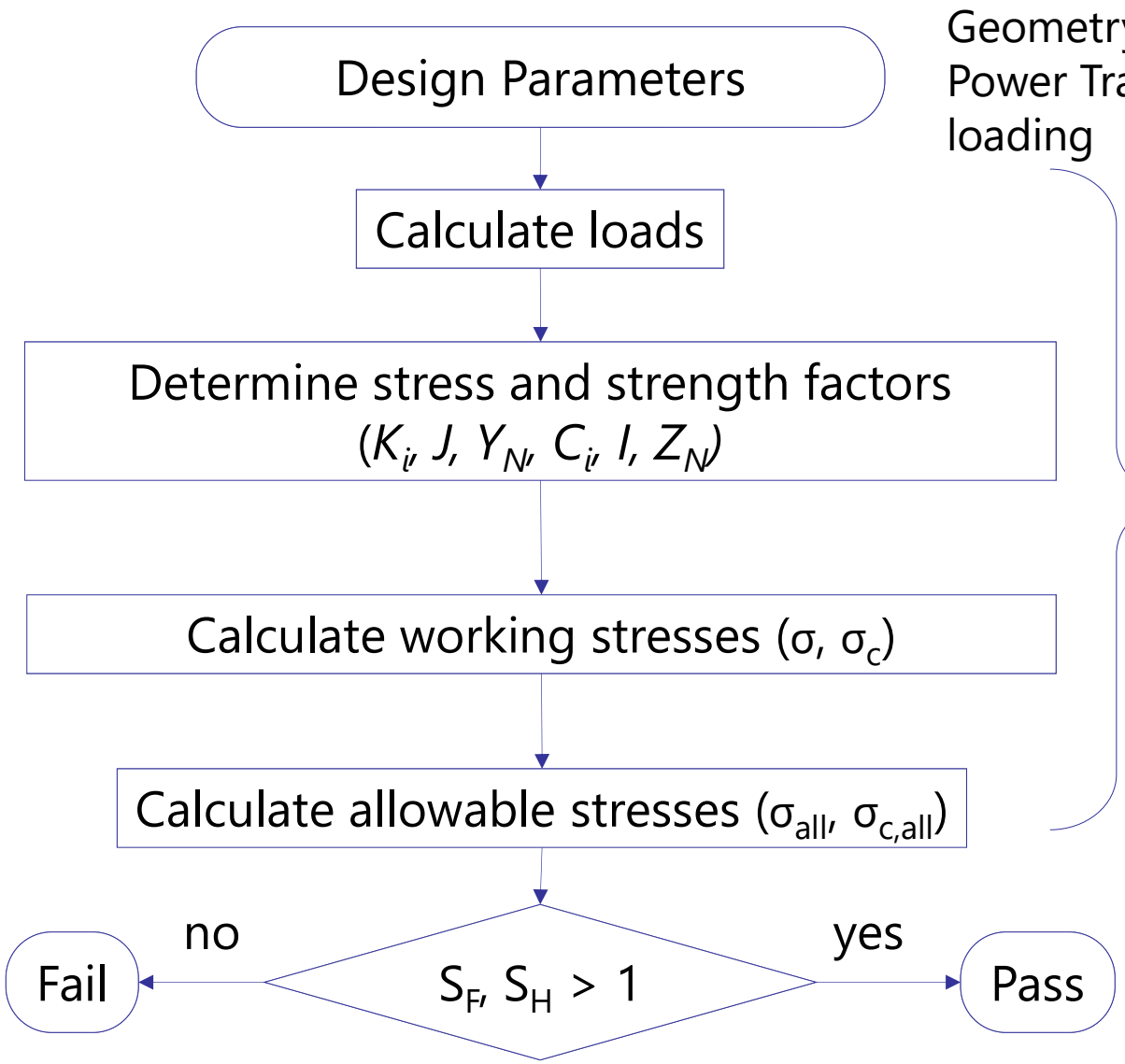


Contact stress safety factor

$$S_H = \frac{S_C}{\sigma_C}\frac{Z_N C_H}{K_T K_R} = n_C$$

- K_T – Temperature Factor (as before)
- K_R – Reliability Factor (as before)

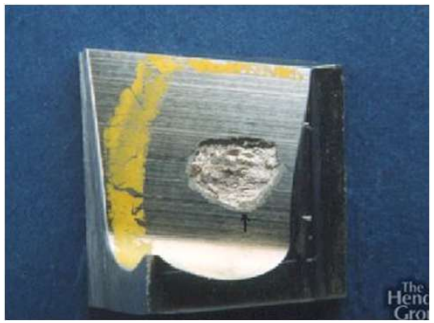
Gear Stress Analysis Procedure



Comparing Bearing and Contact Safety Factors

- Bending stress is linear with transmitted load.
- Contact stress is not linear with transmitted load
- To compare the factors of safety between the different failure modes, to determine which is critical,
  - Compare n_b with n_c^2 for linear or helical contact
  - Compare n_b with n_c^3 for spherical contact





Design of a Gear Mesh - Details

1. A priori decisions (established in advanced)
   - Function: load, speed, reliability, life, overload factor
   - Gear ratio ($m_G = N_G/N_P$)
   - Tooth system: ie. straight vs. helical, addendum, dedendum values, root fillet radius ….
   - Quality number (Q_v)
2. Choose a diametral pitch
   - Initially select median face width $4\pi/P$
   - Find range of necessary ultimate strengths

Design of a Gear Mesh - Details

3. Choose pinion material and core hardness
   - Find face width to meet safety factor in bending
   - Choose face width
   - Check safety factor in bending
   - Find necessary contact fatigue strength, S_c
   - Choose a case hardness
   - Check safety factor in wear
4. Choose a gear material and core hardness
   - Check safety factor in bending
   - Choose a case hardness
   - Check safety factor in wear

AGMA Bending Stress Equations (Mott)

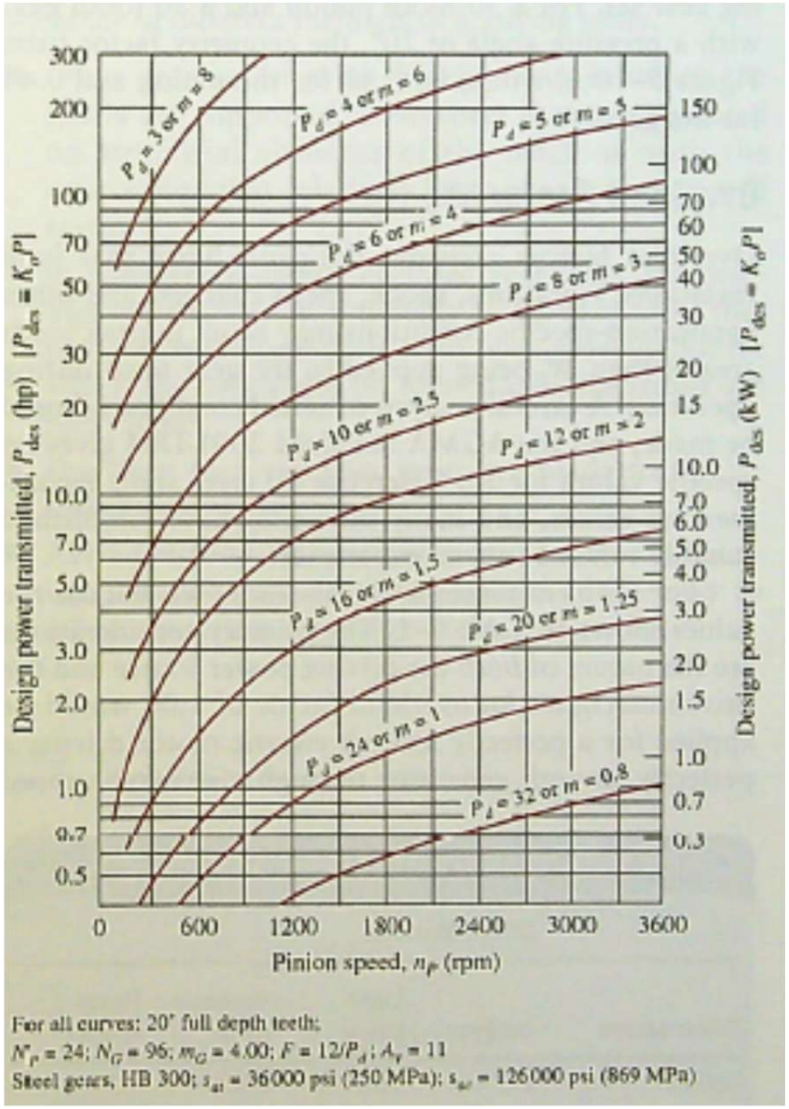
Bending Stress Number, s_t

$$s_t = \frac{W_t P_d}{FJ} K_o K_s K_m K_B K_v \qquad (9\text{–}16)$$

where W_t = tangential force
P_d = diametral pitch
F = face width
J = geometry factor
K_o = overload factor for bending strength
K_s = size factor for bending strength
K_m = load-distrubution factor for bending strength
K_B = rim thickness factor
K_v = dynamic factor for bending strength

- Figure 9-11 provides a good first order approach to estimate P_d given *Design Horsepower* ($K_o H_{nom}$)
- Estimate face width F with

  $8/P_d < F < 12/P_d$



Design of a Gear Mesh - Notes

- Iterate at each stage until no decisions are changed
- If you cannot find a reasonable solution at one stage, you need to move back to the previous stage
- This is not a rigid process; you can work in any sequence
- The process is the similar for all gear types